1 Aeroelastic Flutter

This chapter first provides an introduction to structural wind engineering and extends it to flutter. Next, the flutter mechanism and methods to obtain flutter derivatives are discussed. Then the mathematical formulation for calculating the critical wind speed is provided. Finally, the multi-frequency method is introduced and its workflow is compared with the single-frequency method.

1.1 Wind Effect on Structures

For understanding the effect of wind loads on structures, beginning with Alan G. Davenport's wind loading chain is a standard approach as it provides a framework for breaking down the complex problem of wind loading into a series of five interconnected links (Figure 1).

The first link is wind climate, which deals with the macrometeorological scale of wind. It provides information based on the geographical location, high-return-period phenomena such as natural hazards, and annual wind variations. This information is crucial for determining the maximum wind load that a structure must withstand over its service life.

The second link is the influence of terrain. This step moves from the larger spatial scale of the previous link to the specific location of the building. To understand it, one must first know the atmospheric boundary layer (ABL). The ABL describes the wind profile as a function of height (Figure 2). Due to the viscosity of the wind and the turbulence caused by buildings and the ground, wind velocities near the surface are low but increase with height. This height-dependent profile is defined by the ABL. Importantly, the ABL profile varies depending on the terrain where the structure is located. For example, a building in a dense urban area experiences lower wind velocities at its base compared to a structure near the seashore.

The third link is aerodynamics. This step converts the wind profile, which varies in both time and space, into aerodynamic forces acting on the building. Figures 2 and 3 should be viewed together to appreciate the temporal and spatial variation of wind loads.

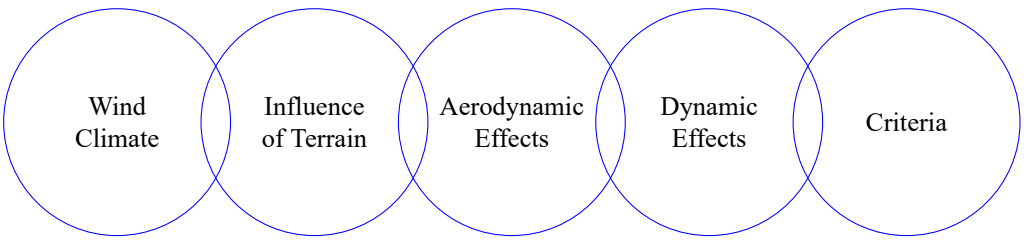


Figure 1: Davenport's wind loading chain

The fourth link is the structural response to these forces. The fifth link is the criteria for assessing the structural performance under these loads. The focus of this thesis is mainly on the 3rd and 4th links (aerodynamics and structural response). Before delving into them, a few points about the Van der Hoven spectrum are discussed, as it provides context into the time scales that links 3 and 4 deal with, relative to wind climate.

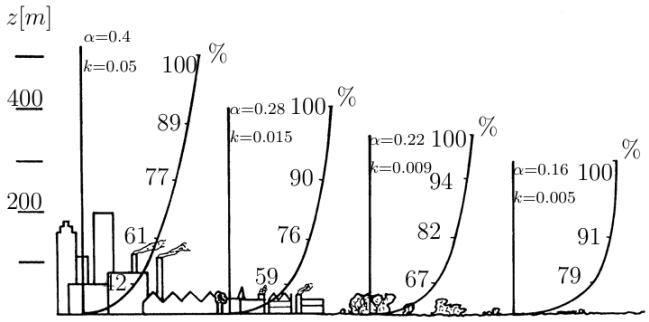


Figure 2: Atmospheric boundary layer wind profiles for different terrain categories [**?**]

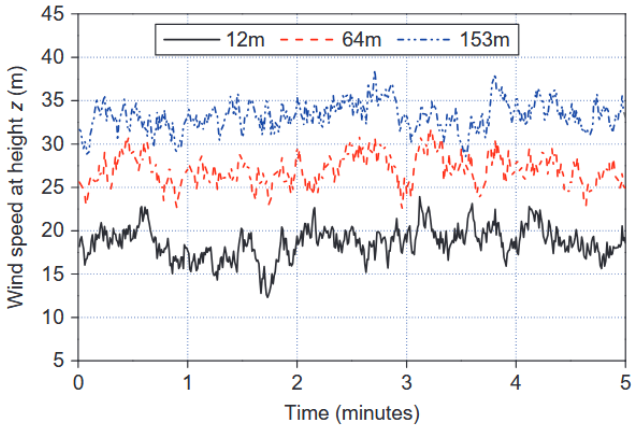


Figure 3: Time history of wind velocity showing turbulent fluctuations [**?**]

In the Van der Hoven spectrum, the x-axis represents frequency and the y-axis represents kinetic energy (Figure 4). This graph reveals which wind cycles are dominant. For instance, a peak can be observed at a period of approximately 0.5 days, corresponding to the daily wind cycle. Another important observation is the presence of a spectral gap, which separates the macrometeorological and micrometeorological characteristics of wind, effectively distinguishing wind climate from aerodynamic forces.

In analysing the forces acting on a structure due to wind, the shape of the structure, along with wind characteristics, strongly affects the aerodynamic forces. Depending on the shape, aerodynamic analysis is classified into two categories: bluff bodies and streamlined bodies. Civil engineering structures mostly fall under the category of bluff body aerodynamics.

A bluff body can be understood as an object that disturbs the streamline flow of a fluid and causes

flow separation at defined points. A wake is the region downstream of the body where the flow is highly disturbed and exhibits negative pressures. When fluid flows over a structure, it separates at the separation points, forming a wake region in the downstream.

The response of a structure to wind load can be broadly classified into two types: along-wind and across-wind responses. Along-wind responses are forces acting on the structure parallel to the wind direction. In across-wind response, forces act perpendicular to the wind direction. Since wind loads vary with time, the structural response to these loads are dynamic in nature. Therefore, the analysis of structures subjected to wind must differentiate between static and dynamic analysis.

In static analysis, the response of the structure to the mean wind load is evaluated. In dynamic analysis, the response due to the fluctuating (time-varying) component of the wind load is considered. Static analysis primarily involves the along-wind response generated by the drag force exerted by the wind on the structure. The major sources of dynamic wind loads are responses due to extraneously induced excitations, instability-induced excitations, and movement-induced excitations. The response of the structure due to these phenomena can be observed in Figure 5.

Buffeting is a primary example of extraneously induced excitation, which acts as an along-wind response caused by wind turbulence. As shown in Figure 3, the mean of the wind velocity (acting on the structure) over time represents the static component, while the variance corresponds to the buffeting component. The calculation of the buffeting component is given in EN 1991-1-4 [**?**].

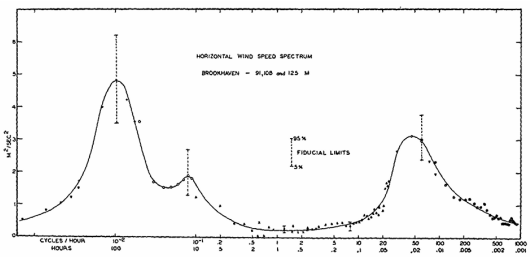


Figure 4: Van der Hoven spectrum of wind kinetic energy [**?**]

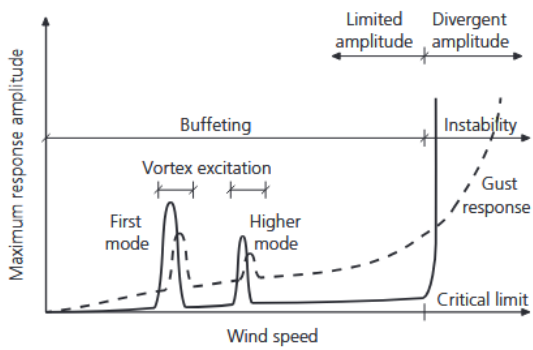


Figure 5: Structural response amplitude versus wind speed under uniform (solid line) and turbulent flow (dashed line) [**?**]

Next are instability-induced excitations. Vortex-induced vibration (VIV) is an example of instability-induced excitation, which is an across-wind response that occurs in a bluff body. This is caused by the alternate shedding of vortices in the wake region of the structure. When the velocity (Reynolds number) of the fluid is low, the vortex is time-invariant, and the structure is subjected to a constant along-wind force, i.e., static and buffeting response. But when the velocity is high, the flow becomes unstable, causing the vortices to shed alternately from the two sides of the structure. This induces an alternating time-varying force of a sinusoidal manner in the across-wind direction. This causes vibration in the structure known as vortex-induced vibration. Since this is a dynamic force,

it requires attention to the calculation of the frequency, as this can determine the resonance of the building. When the frequency of vortex shedding matches the natural frequency of the structure, it causes resonance, leading to large-amplitude vibrations. This can be observed in Figure 5, where a two small local maximum appears in the vortex shedding region. The vortex shedding frequency is given by the formula $f = \mathrm{St}\, U/D$, where D is the characteristic length of the structure and U is the velocity of the fluid. St is the non-dimensional parameter called the Strouhal number, which is a shape constant. For regular shapes, the value of St is taken from EN 1991-1-4.

Finally, movement-induced excitations are caused by the interaction between the wind and the structure. When wind passes over a structure, it creates a force that causes deformation or displacement in the structure. Due to this displacement, the flow around the structure changes and consequently the force acting on the structure changes (also depending on the rate of change of displacement of the structure, the force acting changes). This can create a positive feedback loop where the deformation causes an increase in the force, which in turn causes more deformation. This leads to a rapid increase in the amplitude of vibrations, which can ultimately result in structural failure. These phenomena are known as movement-induced vibrations, as an initial motion is first required to trigger this phenomenon.

Movement-induced excitations mostly cause instability failures in civil engineering structures. This is because once the threshold limit is reached, the structure cannot return to a stable state and fails directly. Flutter and galloping are examples of these types of dynamic instabilities. It can also be observed in Figure 5 that once the flutter limit point is reached, the response rapidly increases. Torsional divergence is also of this type, except that this instability is static in nature. Torsional divergence can be observed in the failure of solar panels, etc. Instability indicates a loss in any one of the structural parameters such as stiffness or damping due to the interaction with wind. In torsional divergence, the stiffness of the structure is lost due to the interaction with wind. This is analogous to structural stability failure like buckling. In buckling, stiffness is lost due to the second-order effect, and in this case it is due to fluid-structure interaction (FSI). Also, similar to buckling, the failure due to this phenomenon depends on the flexibility of the structure rather than the ultimate strength [**?**].

In flutter, the loss is in damping. Since flutter is the focus of this thesis, the following sections focus on discussing the mathematical formulations for quantifying this loss. Before discussing flutter, galloping is discussed first. This is done because galloping is a phenomenon similar to flutter but with less complexity, and hence it can be useful in the intuitive understanding of these dynamic aeroelastic phenomena.

1.2 Galloping

Consider a 1-DOF spring-mass system with a bluff body attached to it (Figure 6). When the wind passes over the bluff body, it creates drag and lift forces. Assume, due to an external forced vibration or VIV, the structure starts to vibrate in the vertical direction. Due to this vertical vibration, depending on the structural velocity, the angle of attack of the wind on the bluff body changes. Since the angle of attack changes, the lift and drag forces also change (except for sections where, for any angle of attack, the lift force remains constant — like a circle). If the change in forces is such that it increases the motion amplitude, then a positive feedback loop is created. This is known as galloping. For this reason, a purely circular rope or cable does not undergo galloping, as the lift force is constant for all angles of attack. But a rectangular section undergoes galloping as the lift force changes with angle of attack.

Next, the analytical formulation for galloping is derived. This formulation is based on the assumption that the instantaneous excitation force acting on the moving structure is nearly equivalent to the static force calculated considering the change in angle of attack due to the velocity of the structure [**?**]. This means there is no phase difference between applied force and structural velocity.

For the system shown in Figure 6(a), the net vertical force acting on the structure is

$$F_y = -F_L \cos\theta - F_D \sin\theta \tag{1.1}$$

where

$$F_L = C_L \frac{\rho d V_{\text{rel}}^2}{2}, \quad F_D = C_D \frac{\rho d V_{\text{rel}}^2}{2}.$$

C_L is the coefficient of lift force, C_D is the coefficient of drag force and ρ_{air} is the density of air.

$$V_{\text{rel}} = \sqrt{V^2 + \dot{y}^2}, \quad \theta = \tan^{-1}\left(\frac{\dot{y}}{V}\right)$$

— relative velocity of wind to the structure's velocity.

Writing the vertical force in terms of C_y

$$F_y = C_y \frac{\rho d V_{\text{rel}}^2}{2}$$

where

$$C_y = -\frac{1}{\cos\theta}(C_L + C_D \tan\theta)$$

The equations of motion for the system can be written as

$$m\,\ddot{y} + c\,\dot{y} + k\,y = F_{\mathrm{y}}(\alpha) = \frac{\rho_{\mathrm{air}}\,V^2}{2}\,d\,C_y(\alpha) \tag{1.2}$$

To solve the equation, the system is linearised by assuming small motion at zero angle of attack

$$\alpha \approx \frac{\dot{y}}{V} \tag{1.3}$$

For this condition

$$F_{\mathrm{y}} \approx \left.\frac{\mathrm{d}F_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \alpha \tag{1.4}$$

which gives the following equations

$$m\,\ddot{y} + c\,\dot{y} + k\,y = \left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \frac{\dot{y}}{V}\frac{\rho\,V^2}{2}\,d \tag{1.5}$$

$$m\,\ddot{y} + \left(c\,-\,\left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \frac{\rho\,V}{2}\,d\right)\dot{y} + k\,y = 0 \tag{1.6}$$

From this equation, it can be seen how the system damping is affected by the interaction with wind: Effective damping = $\left(c\,-\,\left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \frac{\rho\,V}{2}\,d\right)$. When this effective damping becomes negative, the system becomes unstable and galloping occurs. Hence, the critical wind speed for galloping is obtained by setting the effective damping to zero, which yields

$$V_{\mathrm{crit}} = \frac{2c}{\left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0}\rho\,d} \tag{1.7}$$

This formula can be better understood by relating it to the inference that was arrived earlier. For a circle, $\frac{\mathrm{d}C_y}{\mathrm{d}\alpha} = 0$, which makes the entire aerodynamic terms zero [**?**]. To determine whether a structure will undergo galloping, CFD simulations are conducted without fluid-structure interaction (FSI) at various angles of attack. The lift and drag coefficients are extracted from these simulations. Subsequently, the slope of the aerodynamic force function at zero angle of attack is determined and substituted into the critical galloping equation presented above. This approach enables the analysis of structural response subjected to unsteady forces using quasi-steady analysis. This is the result of the assumption introduced earlier in the galloping formulation.

1.3 Flutter Analysis

Flutter is a dynamic instability phenomenon that occurs when the aerodynamic forces acting on a structure couple with its natural modes of vibration, leading to self-excited oscillations. Consider a 2-DOF system of vertical (heave) and rotational (pitch) motion (Figure 7). If the vertical and pitch motions are coupled such that the aerodynamic forces due to the vertical motion cause an increase in the pitch motion, and the aerodynamic forces due to the pitch motion cause an increase

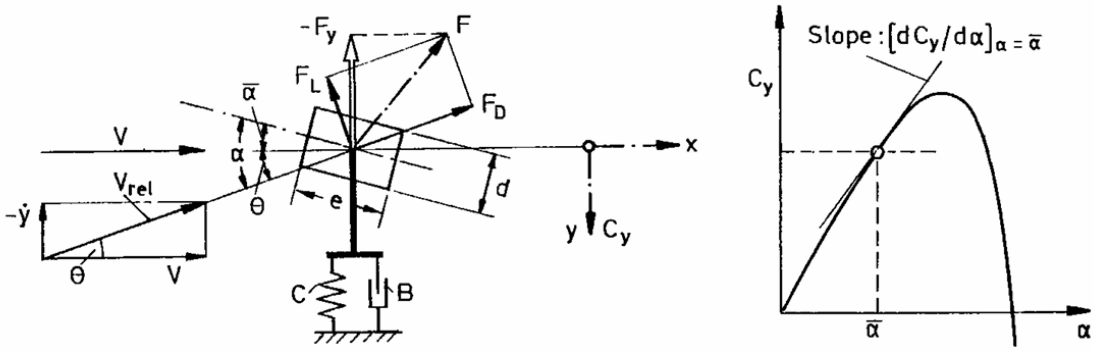


Figure 6: SDOF galloping model and variation of vertical force coefficient with angle of attack [**?**]

in the vertical motion, then a positive feedback loop is created. This can lead to a rapid increase in the amplitude of oscillations, which can ultimately result in structural failure.

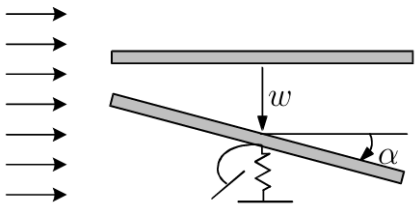


Figure 7: 2-DOF sectional model for flutter analysis (heave and pitch)

This is illustrated in Figure 8. In the Figure 8a, it can be observed that the phase between the vertical and pitch motion is such that the positive work created during the upward motion is cancelled by the negative work created during the downward motion. Hence, the system is stable. But in the Figure 8b, the phase between the vertical and pitch motion is such that positive work is created during both the upward and downward motion. Hence, a positive feedback loop is created and the system becomes unstable. One can refer to the work of Billah et.al [**?**], where a detailed discussion on above mentioned concepts in the context of failure of Tacoma Narrows Bridge is presented.

Flutter analysis, like galloping, involves determining the critical wind speed at which structural instability initiates. However, unlike galloping (which is a single-degree-of-freedom system), flutter analysis considers two degrees of freedom and includes phase differences between the applied aerodynamic forces and the structural response. Consequently, flutter analysis must account for unsteady aerodynamic forces, making it considerably more complex than galloping.

The dynamic equilibrium equations for a 2-DOF system are

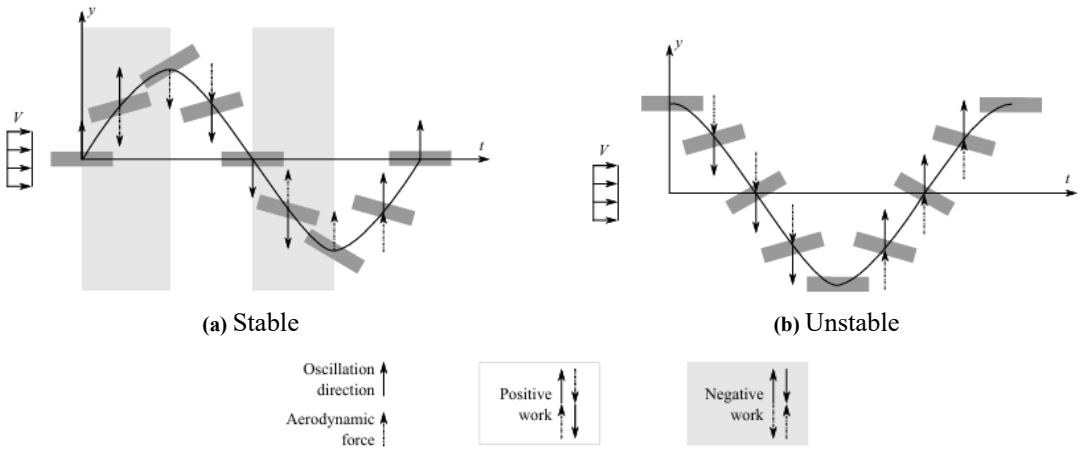


Figure 8: Flutter mechanism: stable versus unstable phase relationships [**?**]

$$m(\ddot{h} + 2\omega_h \xi_h \dot{h} + \omega_h^2 h) = L_{ae} \tag{1.8}$$

$$I_\alpha(\ddot{\alpha} + 2\omega_\alpha \xi_\alpha \dot{\alpha} + \omega_\alpha^2 \alpha) = M_{ae} \tag{1.9}$$

L_{ae} is the aerodynamic lift force due to wind and M_{ae} is the aerodynamic moment due to wind. In these equations, quantifying unsteady aerodynamic forces is the challenge. In 1934, Theodorsen, a Norwegian-American aerodynamicist, developed an analytical formulation for the aerodynamic forces for unsteady flutter analysis of an airfoil [**?**]. The solutions for L_{ae} and M_{ae} are functions of Theodorsen circulatory functions, which, in turn, are functions of K, the reduced frequency, where $K = \frac{\omega B}{U}$. ω is the oscillating frequency in radians per second, B is the width of the cross-section and U is the velocity of the fluid. The reduced frequency can be understood as the oscillating velocity relative to the fluid flow velocity. This quantifies the time lag between the applied force and the structure's displacement. As this theory was based on the assumption of potential flow, it cannot be applied to bridges. In 1971, Scanlan, an American civil engineer, reformulated Theodorsen's theory to accommodate bridges using a semi-empirical method based on eight functions (for a 2-DOF system) known as flutter derivatives [**?**].

$$L_{ae} = \frac{1}{2}\rho U^2 B \left[K H_1^* \frac{\dot{h}}{U} + K H_2^* \frac{B\dot{\alpha}}{U} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B} \right] \tag{1.10}$$

$$M_{ae} = \frac{1}{2}\rho U^2 B^2 \left[K A_1^* \frac{\dot{h}}{U} + K A_2^* \frac{B\dot{\alpha}}{U} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B} \right] \tag{1.11}$$

In these equations, h denotes the vertical degree of freedom (called heave motion) and α denotes the rotational degree of freedom (pitch motion). The quantities H_1^*, H_2^*, H_3^*, H_4^*, A_1^*, A_2^*, A_3^*, A_4^* are the flutter derivatives. These are shape constants that quantify the wind force acting on a structure as a linear relation of the section's vertical velocity, angular rotation, and speed of angular rotation. They are in turn functions of $U_{red} = \frac{1}{K} = \frac{U}{fB}$. These derivatives can be understood as

follows: just as C_d is a non-dimensional shape constant for computing drag force on a structure, each flutter derivative quantifies a particular aeroelastic lift or moment component. For example, H_1^* is the flutter derivative for quantifying the lift force due to vertical velocity of the structure. A_3^* is the flutter derivative for quantifying the moment force due to angular rotation of the structure.

Since this is a semi-empirical method, these derivatives must be found through CFD simulations or wind-tunnel tests. Once obtained, they are applied back to the dynamic equilibrium equations to determine the critical velocity. The approaches for finding the flutter derivatives can be broadly classified based on the nature of the applied motion: forced motion tests or free vibration. In forced motion tests, the structure is forced to oscillate at a specific frequency and amplitude, and the resulting aerodynamic forces are measured to determine the flutter derivatives. In free vibration tests, the structure is allowed to vibrate freely under the influence of aerodynamic forces, and the flutter derivatives are determined from the observed damping response. For the preliminary analysis of flutter, forced motion tests with CFD have greater importance due to their easier setup and lower cost. Since forced motion tests are the focus of this thesis, only this approach is explained in the following sections.

1.4 Identification of Flutter Derivatives with Forced Motion Simulations

A 2-D cross-section of the bridge deck is modelled in a virtual wind tunnel with appropriate boundary conditions. The bridge cross-section is then oscillated sinusoidally in the vertical (heave) direction according to

$$h(t) = h_0 \sin(\omega t), \tag{1.12}$$

$$\dot{h}(t) = \omega h_0 \cos(\omega t), \tag{1.13}$$

where h_0 is the maximum amplitude of the applied heave motion and ω is the angular frequency of the applied heave motion. Since no pitch motion is applied, $\alpha = \dot{\alpha} = 0$, and the aerodynamic lift force (equation 1.10) and pitching moment (equation 1.11) simplify to

$$L_{ae} = \frac{1}{2}\rho U^2 B \left[K H_1^* \frac{\dot{h}}{U} + K^2 H_4^* \frac{h}{B} \right] \qquad \text{(Scanlan formulation)}, \tag{1.14}$$

$$M_{ae} = \frac{1}{2}\rho U^2 B^2 \left[K A_1^* \frac{\dot{h}}{U} + K^2 A_4^* \frac{h}{B} \right] \qquad \text{(Scanlan formulation)}. \tag{1.15}$$

Under the small-amplitude assumption (small h_0), Scanlan's linearisation implies that the aerodynamic lift and moment measured from CFD take the sinusoidal form

$$L_{ae} = B_0 \sin(\omega t + \phi) \;=\; B_1 \cos(\omega t) + B_2 \sin(\omega t), \tag{1.16}$$

$$M_{ae} = C_0 \sin(\omega t + \Phi) \;=\; C_1 \cos(\omega t) + C_2 \sin(\omega t). \tag{1.17}$$

The flutter derivatives are determined by equating the lift force from the Scanlan formulation with the CFD result and matching cosine and sine coefficients, which gives

$$\frac{1}{2}\rho U^2 B \left[K H_1^* \omega h_0 \cos(\omega t) + K^2 H_4^* h_0 \sin(\omega t)\right] = B_1 \cos(\omega t) + B_2 \sin(\omega t), \tag{1.18}$$

$$\frac{1}{2}\rho U^2 B^2 \left[K A_1^* \omega h_0 \cos(\omega t) + K^2 A_4^* h_0 \sin(\omega t)\right] = C_1 \cos(\omega t) + C_2 \sin(\omega t). \tag{1.19}$$

From this, the flutter derivatives for heave excitation are obtained as

$$H_1^* = \frac{2B_1}{\rho\,\omega^2 B^2 h_0}, \qquad H_4^* = \frac{2B_2}{\rho\,\omega^2 B^2 h_0}, \tag{1.20}$$

$$A_1^* = \frac{2C_1}{\rho\,\omega^2 B^3 h_0}, \qquad A_4^* = \frac{2C_2}{\rho\,\omega^2 B^3 h_0}. \tag{1.21}$$

The same procedure is repeated with pitch (torsional) excitation to obtain the H_2^*, H_3^*, A_2^*, and A_3^* derivatives.

$$H_2^* = \frac{2B_1}{\rho\omega^2 B^3 \theta_0}, \qquad H_3^* = \frac{2B_2}{\rho\omega^2 B^3 \theta_0}, \tag{1.22}$$

$$A_2^* = \frac{2C_1}{\rho\omega^2 B^4 \theta_0}, \qquad A_3^* = \frac{2C_2}{\rho\omega^2 B^4 \theta_0}. \tag{1.23}$$

Consequently, a total of two forced-motion simulations (one heave, one pitch) are required to obtain a complete set of eight flutter derivatives at each reduced velocity.

This is the single-frequency method, where a heave/pitch motion of one frequency is applied to the structure to find the flutter derivatives. Obtaining flutter derivatives with this approach is highly computationally costly for the following reasons. Since flutter derivatives are functions of reduced velocity, multiple CFD simulations (by varying the frequency of the input forced motion) must be performed to obtain them. To obtain each point in a set of eight flutter derivatives, two CFD simulations are required. To obtain an accurate representation of each flutter derivative function, a minimum of six to eight points on the flutter derivative curve is needed. Hence, a total of 16 simulations are required. Each CFD simulation takes approximately 48 hours. Therefore, to obtain a complete set of flutter derivatives, 768 hours of simulation time are required.

1.5 Multi-Frequency Method

In the multi-frequency method, instead of applying a single-frequency forced-motion oscillation at a time and extracting four flutter derivatives, several frequencies are superimposed and applied simultaneously to the structure. In this way, $N \times 4$ flutter derivatives can be obtained from N simulations. For example, if four frequencies are applied simultaneously, then 16 flutter derivatives can be obtained from a single simulation. The superimposition of multiple frequencies can be expressed as

$$h(t) = \sum_{k=1}^{N} h_{0,k} \sin(\omega_k t + \phi_k). \tag{1.24}$$

Under the assumption of aerodynamic linearity, the total aerodynamic response is the superposition of the responses at each individual frequency. Consequently, N sets of flutter derivatives can be extracted from a single simulation by applying a Fourier decomposition of the measured aerodynamic forces at each frequency component ω_k.

The procedures for calculating the critical wind speed (Section 1.6) is identical for both the single and multi-frequency methods. The only difference lies in the identification of aerodynamic forces (Section 1.4), where multiple frequency components are excited simultaneously rather than one at a time. Figures 9 and 10 illustrate the workflow for the two methods.

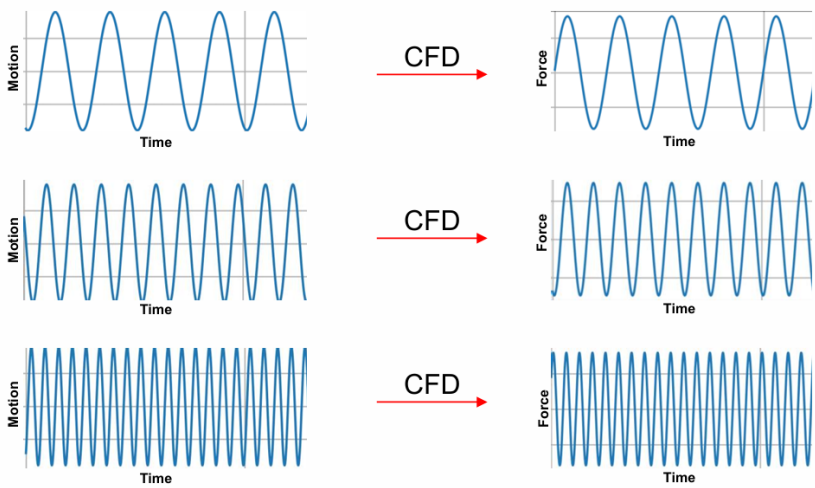


Figure 9: Workflow of the single-frequency forced-motion method

1.6 Calculation of Critical Wind Speed

To obtain the critical wind speed, the equations of motion for the bridge deck (equations 1.8 and 1.9) are equated to the aerodynamic forces from the Scanlan formulation (equations 1.10 and 1.11) and solved. Equating them yields

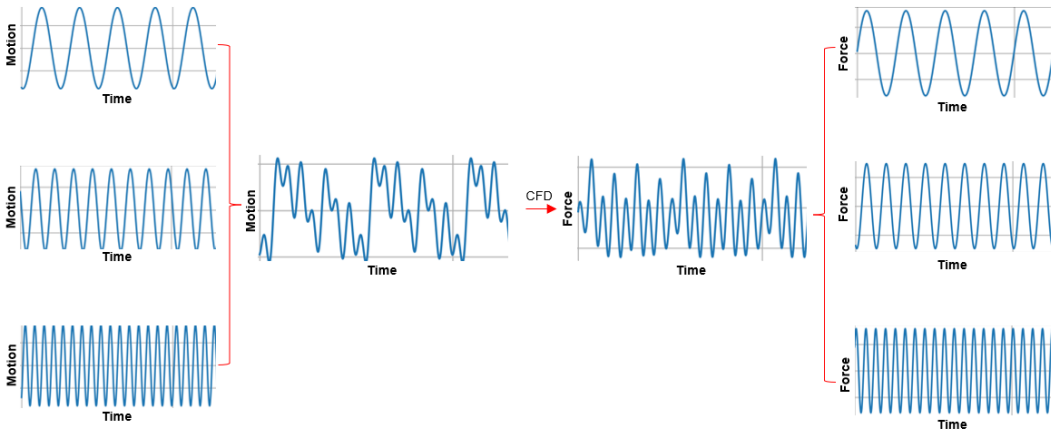


Figure 10: Workflow of the multi-frequency forced-motion method

$$m\left(\ddot{h} + 2\omega_h\zeta_h\dot{h} + \omega_h^2 h\right) = \frac{1}{2}\rho U^2 B\left[KH_1^*\frac{\dot{h}}{U} + KH_2^*\frac{B\dot{\alpha}}{U} + K^2H_3^*\alpha + K^2H_4^*\frac{h}{B}\right], \quad (1.25)$$

$$I_\alpha\left(\ddot{\alpha} + 2\omega_\alpha\zeta_\alpha\dot{\alpha} + \omega_\alpha^2\alpha\right) = \frac{1}{2}\rho U^2 B^2\left[KA_1^*\frac{\dot{h}}{U} + KA_2^*\frac{B\dot{\alpha}}{U} + K^2A_3^*\alpha + K^2A_4^*\frac{h}{B}\right]. \quad (1.26)$$

Converting this to matrix format and rearranging the terms gives

$$\begin{aligned}&\begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix}\begin{Bmatrix} \ddot{h} \\ \ddot{\alpha} \end{Bmatrix} + \begin{bmatrix} C_h & 0 \\ 0 & C_\alpha \end{bmatrix}\begin{Bmatrix} \dot{h} \\ \dot{\alpha} \end{Bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix}\begin{Bmatrix} h \\ \alpha \end{Bmatrix} \\ &- \frac{1}{2}\rho B^2\omega\left(\begin{bmatrix} H_1^* & BH_2^* \\ BA_1^* & B^2A_2^* \end{bmatrix}\begin{Bmatrix} \dot{h} \\ \dot{\alpha} \end{Bmatrix} + \omega^2\begin{bmatrix} H_4^* & BH_3^* \\ BA_4^* & B^2A_3^* \end{bmatrix}\begin{Bmatrix} h \\ \alpha \end{Bmatrix}\right) = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}.\end{aligned} \quad (1.27)$$

This equation is in the time domain. To solve it, harmonic motion $h = h_0 e^{i\omega t}$ and $\alpha = \alpha_0 e^{i\omega t}$ is assumed, and the equation is transformed to the frequency domain. The system becomes

$$\begin{aligned}&\left(-\omega^2\begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix} + i\omega\begin{bmatrix} C_h & 0 \\ 0 & C_\alpha \end{bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix}\right. \\ &\left. -\frac{1}{2}\rho B^2\omega^2\begin{bmatrix} H_4^* - H_1^* & BH_3^* - BH_2^* \\ BA_4^* - BA_1^* & B^2A_3^* - B^2A_2^* \end{bmatrix}\right)\begin{Bmatrix} h_0 \\ \alpha_0 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix},\end{aligned} \quad (1.28)$$

which is of the standard form

$$-\omega^2 M + i\omega C + K = 0. \quad (1.29)$$

The unknown in the equation is $\omega = \omega_r + i\omega_i$, which represents the frequency in complex form. The meaning of frequency in complex form can be interpreted from structural dynamics theory. The solution of a simple harmonic oscillation without damping can be represented as

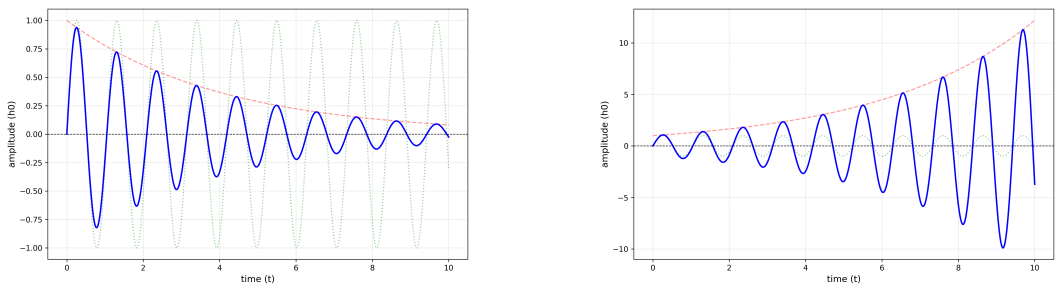
$x(t) = e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$. This represents a pure sinusoidal oscillation with frequency ω (real number) and with a phase angle. To represent the effect of damping, the solution takes the form $x(t) = e^{i\omega t} = e^{i\omega_r t} \cdot e^{-\omega_i t} = e^{-\omega_i t}\big(\cos(\omega t) + i\sin(\omega t)\big)$, where the real part ω_r is the oscillation frequency and the imaginary part ω_i governs the damping of the response. This equation must be solved for ω to determine the system's oscillating frequency and damping characteristics. Because structural damping is present, the problem takes the form of a quadratic eigenvalue problem. Hence, this is linearised as shown below to be solved as a generalised eigenvalue problem:

$$\left(\begin{bmatrix} 0 & K \\ K & iC \end{bmatrix} - \omega \begin{bmatrix} K & 0 \\ 0 & -M \end{bmatrix}\right)\begin{pmatrix} X \\ \omega X \end{pmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \qquad \text{i.e.} \quad A - \lambda B = 0. \tag{1.30}$$

Based on the explanation provided above for damping, the system stability or instability can be determined. For example, consider a solution of $\omega = \omega_r + i\omega_i$, where both are positive. Then, $e^{-\omega_i t}\big(\cos(\omega t) + i\sin(\omega t)\big)$ represents a damped oscillation with frequency ω_r and an amplitude that decays exponentially over time owing to the positive ω_i. If $\omega = \omega_r - i\omega_i$, then $e^{+\omega_i t}\big(\cos(\omega t) + i\sin(\omega t)\big)$ represents the same oscillation but with an amplitude that grows exponentially over time due to the negative ω_i. Thus, the sign of ω_i determines stability:

- $\omega_i > 0$: $\quad h = h_0\, e^{i\omega_r t} \cdot e^{-\omega_i t}$ — stable (amplitude decays).
- $\omega_i < 0$: $\quad h = h_0\, e^{i\omega_r t} \cdot e^{+|\omega_i| t}$ — unstable (amplitude grows; flutter onset).

This is illustrated in Figure 11.



(a) Damped solution

(b) Divergent solution

Figure 11: Stable (damped) and unstable (diverging) time-domain flutter responses

Determining the critical wind speed is an iterative process because the flutter derivatives are functions of reduced velocity, which in turn is a function of wind speed. Hence the following steps (which form the algorithm of the code) are performed iteratively to find the critical wind speed:

1. A reduced velocity $U_r = U/(fB)$ is selected.
2. Flutter derivatives corresponding to that reduced velocity are interpolated from the flutter-derivative functions.

3. The linearised eigenvalue problem is solved for ω.

4. If the system is stable, the reduced velocity is increased, and the process repeats until the system first becomes unstable. The reduced critical wind speed is the speed at which ω_i changes sign from positive to negative, indicating the onset of flutter instability.

5. The actual critical wind speed is then recovered from

$$U_c = U_{r,c} \frac{\omega_r}{2\pi} B, \tag{1.31}$$

where $U_{r,c}$ is the reduced critical velocity and ω_r is the real part of the eigenfrequency at that point.

1.7 Validation of the Critical Velocity Tool

To perform the above-mentioned calculations, a Python tool is developed and validated against the benchmark results of [**?**]. The structural parameters taken for the validation are taken from [**?**] and given in Table 1.

Table 1: Structural parameters used for validation [**?**].

Symbol	Description	Value
m_L	Mass per unit length	$22\,740$ kg/m
J_L	Moment of inertia per unit length	2.47×10^6 kg m^2/m
B	Deck chord (width)	31 m
f_z	Vertical structural eigenfrequency	0.10 Hz
f_θ	Torsional structural eigenfrequency	0.278 Hz
ξ	Damping ratio (both modes)	0.003 $(-)$

Theodorsen derivatives are used for the aerodynamic properties of the cross-section. The Theodorsen flutter derivatives are analytical functions derived from the closed-form solution for a thin flat plate in potential flow, first obtained by Theodorsen [**?**]. They are expressed in terms of the real and imaginary parts of the Theodorsen circulatory function $C(k) = F(k) + iG(k)$, where k is the reduced frequency. The analytical expressions are listed in Table 2. The results obtained with the Python tool is shown in Figure 12.

Since the bridge deck is modelled as a two-degree-of-freedom system, two eigenmodes are obtained. The zero-crossing of the imaginary part of each eigenfrequency is checked; the lowest reduced velocity at which a zero crossing occurs is taken as the reduced critical velocity $U_{r,c}$. Substituting into Equation (1.31) yields

Table 2: Analytical Theodorsen flutter derivative expressions for a flat plate.

	H_i^*	A_i^*
$i=1$	$H_1^* = -\dfrac{2\pi}{K}F$	$A_1^* = \dfrac{\pi}{2K}F$
$i=2$	$H_2^* = -\dfrac{\pi}{2K}\left(1+F+\dfrac{4G}{K}\right)$	$A_2^* = -\dfrac{\pi}{8K}\left(1-F-\dfrac{4G}{K}\right)$
$i=3$	$H_3^* = -\dfrac{2\pi}{K^2}\left(F-\dfrac{KG}{4}\right)$	$A_3^* = \dfrac{\pi}{2K^2}\left(\dfrac{K^2}{32}+F-\dfrac{KG}{4}\right)$
$i=4$	$H_4^* = \dfrac{\pi}{2}\left(1+\dfrac{4G}{K}\right)$	$A_4^* = -\dfrac{\pi}{2K}G$

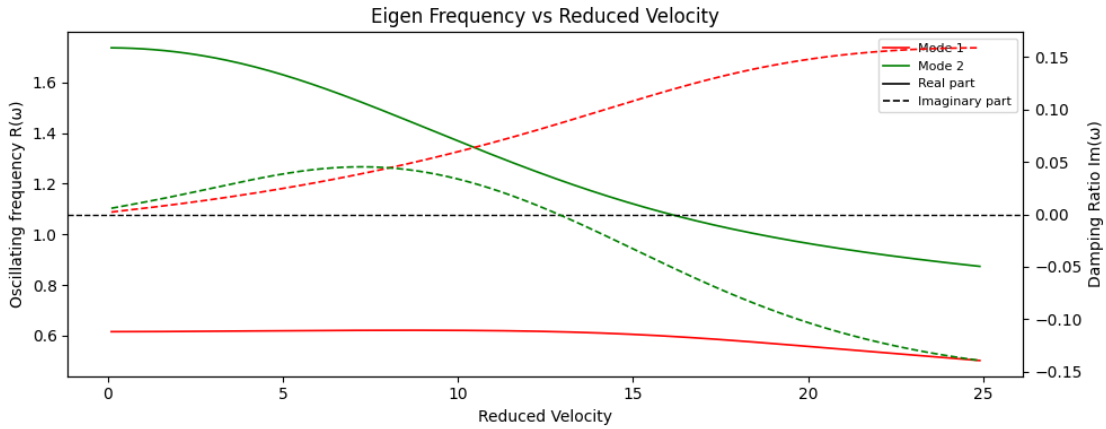


Figure 12: Real and imaginary parts of the eigenfrequency variation against reduced velocity

$$
\begin{aligned}
U_{r,c} &= 12.9, \\
\omega_r &= 1.21 \text{ rad/s}, \\
U_c &= 77.32 \text{ m/s}.
\end{aligned}
$$

This result falls within the reference range of $[77.1,\ 78.6]$ m/s, which is consistent with the range reported by [**?**].